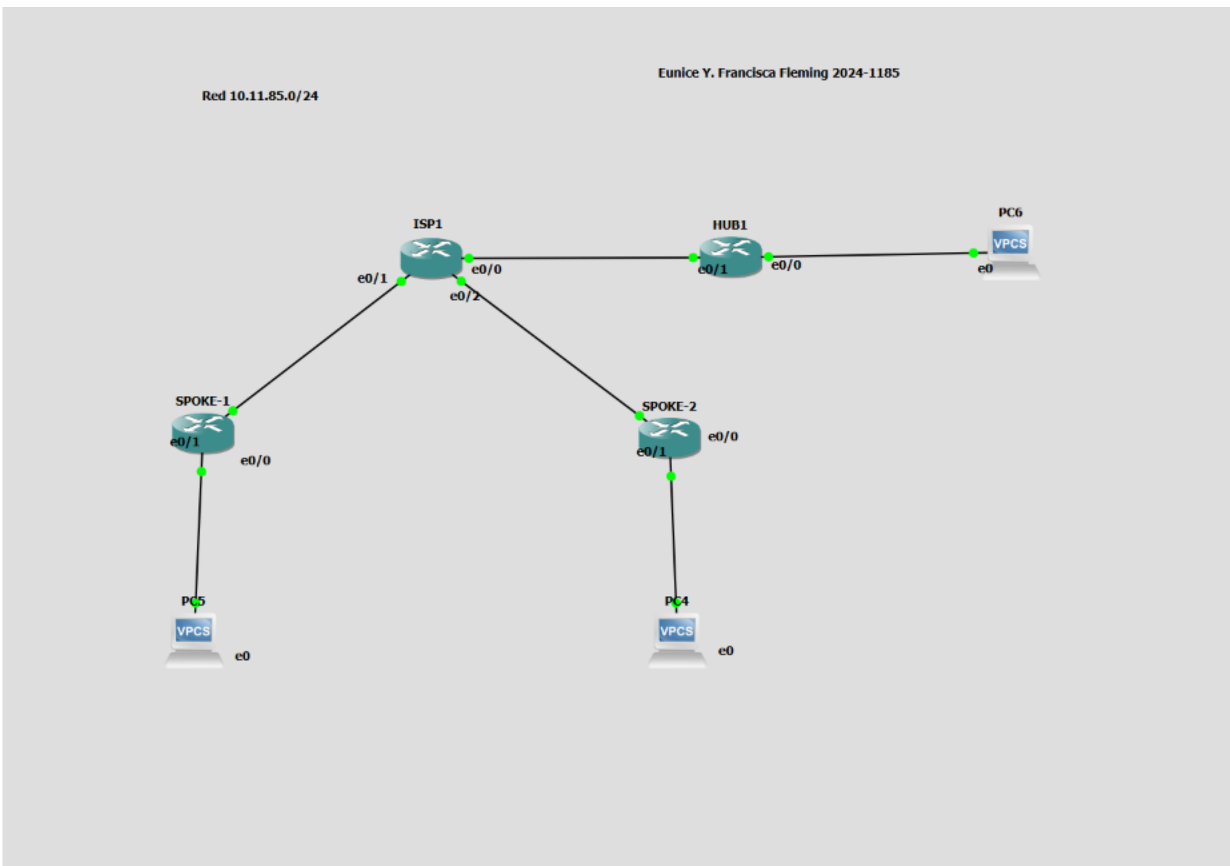
Objetivo

Implementar una VPN hub-and-spoke punto a multipunto DMVPN Fase 3 utilizando IPSec IKEv2 y enrutamiento dinámico EIGRP. Fase 3 introduce NHRP redirect/shortcut: el HUB detecta tráfico subóptimo entre dos spokes y les indica que negocien un túnel directo, actualizando dinámicamente el next-hop en la tabla de rutas. Esto resuelve la limitación de Fase 2, donde el next-hop seguía siendo el HUB incluso tras formar un túnel spoke-to-spoke.

Topología de Red



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
HUB	e0/1	10.11.85.98/30	WAN → ISP

HUB	e0/0	10.11.85.65/27	LAN → PC3
SPOKE1	e0/0	10.11.85.102/30	WAN → ISP
SPOKE1	e0/1	10.11.85.1/27	LAN → PC2
SPOKE2	e0/0	10.11.85.106/30	WAN → ISP
SPOKE2	e0/1	10.11.85.33/27	LAN → PC1
ISP	e0/0	10.11.85.97/30	Hacia HUB
ISP	e0/1	10.11.85.101/30	Hacia SPOKE1
ISP	e0/2	10.11.85.105/30	Hacia SPOKE2

Parámetros

IKEv2

Parámetro	Valor
Proposal	PROP-DMVPN
Cifrado	AES-CBC-256
Integridad	SHA-256
Grupo DH	14 (2048-bit)
Keyring	KR-DMVPN (wildcard en HUB)
Profile	PROF-DMVPN-IKEV2
PSK	ITLA2024

IPSec / NHRP / Túnel

IPSec Profile	PROF-DMVPN-VTI (+ set ikev2-profile)
Tunnel mode	gre multipoint
NHRP redirect (HUB)	Habilitado
NHRP shortcut (Spokes)	Habilitado

EIGRP

Parámetro	Valor
AS Number	1
Auto-summary	Deshabilitado
Extra en HUB	no ip split-horizon eigrp 1 + no ip next-hop-self eigrp 1

Scripts de Configuración

ISP

enable
configure terminal
hostname ISP1

interface Ethernet0/0
description Hacia HUB
ip address 10.11.85.97 255.255.255.252
no shutdown

interface Ethernet0/1

Parámetro	Valor
Transform Set	TS-DMVPN-IKEV2
Modo	Transport

description Hacia SPOKE1
ip address 10.11.85.101 255.255.255.252
no shutdown

interface Ethernet0/2
description Hacia SPOKE2

```
 ip address 10.11.85.105 255.255.255.252
 no shutdown

ip route 10.11.85.64 255.255.255.224 10.11.85.98
ip route 10.11.85.0  255.255.255.224 10.11.85.102
ip route 10.11.85.32 255.255.255.224 10.11.85.106

end
```

HUB

```
enable
configure terminal

hostname HUB1

interface Ethernet0/1
 description WAN hacia ISP
 ip address 10.11.85.98 255.255.255.252
 no shutdown

interface Ethernet0/0
 description LAN hacia PC3
 ip address 10.11.85.65 255.255.255.224
 no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.97

crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14

crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN


crypto ikev2 keyring KR-DMVPN
 peer ANY-SPOKE
  address 0.0.0.0 0.0.0.0
  pre-shared-key ITLA2026
 !


crypto ikev2 profile PROF-DMVPN-IKEV2
```

```
 match identity remote address 0.0.0.0
 authentication remote pre-share
 authentication local pre-share
 keyring local KR-DMVPN

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
 mode transport

crypto ipsec profile PROF-DMVPN-VTI
 set transform-set TS-DMVPN-IKEV2
 set ikev2-profile PROF-DMVPN-IKEV2

interface Tunnel0
 description DMVPN Fase3 - Hub
 ip address 10.10.10.1 255.255.255.224
 no ip redirects
 ip mtu 1400
 ip nhrp authentication ITLA-DMVPN
 ip nhrp map multicast dynamic
 ip nhrp network-id 85185
 ip nhrp holdtime 300
 ip nhrp redirect
 no ip split-horizon eigrp 1
 no ip next-hop-self eigrp 1
 tunnel source Ethernet0/1
 tunnel mode gre multipoint
 tunnel key 85185
 tunnel protection ipsec profile PROF-DMVPN-VTI
 no shutdown

router eigrp 1
 no auto-summary
 network 10.10.10.0 0.0.0.31
 network 10.11.85.64 0.0.0.31

end
```

SPOKE1

```
enable
configure terminal

hostname SPOKE1
interface Ethernet0/0
```

```
 description WAN hacia ISP
 ip address 10.11.85.102 255.255.255.252
 no shutdown

interface Ethernet0/1
 description LAN hacia PC2
 ip address 10.11.85.1 255.255.255.224
 no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.101
crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14

crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN

crypto ikev2 keyring KR-DMVPN
 peer HUB
  address 10.11.85.98
  pre-shared-key ITLA2026
 !

crypto ikev2 profile PROF-DMVPN-IKEV2
 match identity remote address 10.11.85.98 255.255.255.255
 authentication remote pre-share
 authentication local pre-share
 keyring local KR-DMVPN

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
 mode transport
crypto ipsec profile PROF-DMVPN-VTI
 set transform-set TS-DMVPN-IKEV2
 set ikev2-profile PROF-DMVPN-IKEV2

interface Tunnel0
 description DMVPN Fase3 - Spoke1
 ip address 10.10.10.2 255.255.255.224
 no ip redirects
 ip mtu 1400
 ip nhrp authentication ITLA-DMVPN
 ip nhrp map 10.10.10.1 10.11.85.98
 ip nhrp map multicast 10.11.85.98
```

```
 ip nhrp network-id 85185
 ip nhrp holdtime 300
 ip nhrp nhs 10.10.10.1
 ip nhrp shortcut
 tunnel source Ethernet0/0
 tunnel mode gre multipoint
 tunnel key 85185
 tunnel protection ipsec profile PROF-DMVPN-VTI
 no shutdown

router eigrp 1
 no auto-summary
 network 10.10.10.0 0.0.0.31
 network 10.11.85.0 0.0.0.31

end
```

SPOKE2

```
enable
configure terminal

hostname SPOKE2

interface Ethernet0/0
 description WAN hacia ISP
 ip address 10.11.85.106 255.255.255.252
 no shutdown

interface Ethernet0/1
 description LAN hacia PC1
 ip address 10.11.85.33 255.255.255.224
 no shutdown

ip route 0.0.0.0 0.0.0.0 10.11.85.105

crypto ikev2 proposal PROP-DMVPN
 encryption aes-cbc-256
 integrity sha256
 group 14

crypto ikev2 policy POL-DMVPN
 proposal PROP-DMVPN
crypto ikev2 keyring KR-DMVPN
```

```
 peer HUB
  address 10.11.85.98
  pre-shared-key ITLA2026
 !
crypto ikev2 profile PROF-DMVPN-IKEV2
 match identity remote address 10.11.85.98 255.255.255.255
 authentication remote pre-share
 authentication local pre-share
 keyring local KR-DMVPN

crypto ipsec transform-set TS-DMVPN-IKEV2 esp-aes 256 esp-sha256-hmac
 mode transport

crypto ipsec profile PROF-DMVPN-VTI
 set transform-set TS-DMVPN-IKEV2
 set ikev2-profile PROF-DMVPN-IKEV2
interface Tunnel0
 description DMVPN Fase3 - Spoke2
 ip address 10.10.10.3 255.255.255.224
 no ip redirects
 ip mtu 1400
 ip nhrp authentication ITLA-DMVPN
 ip nhrp map 10.10.10.1 10.11.85.98
 ip nhrp map multicast 10.11.85.98
 ip nhrp network-id 85185
 ip nhrp holdtime 300
 ip nhrp nhs 10.10.10.1
 ip nhrp shortcut
 tunnel source Ethernet0/0
 tunnel mode gre multipoint
 tunnel key 85185
 tunnel protection ipsec profile PROF-DMVPN-VTI
 no shutdown
router eigrp 1
 no auto-summary
 network 10.10.10.0 0.0.0.31
 network 10.11.85.32 0.0.0.31

end
```

VPCS

```
! PC4
ip 10.11.85.34 255.255.255.224 10.11.85.33
save
```

! PC5
ip 10.11.85.2 255.255.255.224 10.11.85.1
save

! PC6
ip 10.11.85.66 255.255.255.224 10.11.85.65
save

Verificación

```
show ip route static
```

```
ISP1#show ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
S        10.11.85.0/27 [1/0] via 10.11.85.102
S        10.11.85.32/27 [1/0] via 10.11.85.106
S        10.11.85.64/27 [1/0] via 10.11.85.98
ISP1#
```

```
show crypto ikev2 sa
```

```
SPOKE1#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         10.11.85.102/500      10.11.85.98/500       none/none            READY
      Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/1815 sec

 IPv6 Crypto IKEv2  SA
```

```
HUB1#show crypto ikev2 sa
 IPv4 Crypto IKEv2  SA

Tunnel-id Local                 Remote                fvrf/ivrf            Status
2         10.11.85.98/500       10.11.85.102/500      none/none            READY
      Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/1783 sec

Tunnel-id Local                 Remote                fvrf/ivrf            Status
1         10.11.85.98/500       10.11.85.106/500      none/none            READY
      Encr: AES-CBC, keysize: 256, Hash: SHA256, DH Grp:14, Auth sign: PSK, Auth verify: PSK
      Life/Active Time: 86400/1790 sec

 IPv6 Crypto IKEv2  SA
```

show ip nhrp

```
HUB1#show ip nhrp
10.10.10.2/32 via 10.10.10.2
   Tunnel0 created 00:31:01, expire 00:04:39
   Type: dynamic, Flags: unique registered used nhop
   NBMA address: 10.11.85.102
10.10.10.3/32 via 10.10.10.3
   Tunnel0 created 00:31:08, expire 00:03:23
   Type: dynamic, Flags: unique registered used nhop
   NBMA address: 10.11.85.106
HUB1#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.11.85.102         10.10.10.2    UP 00:34:25     D
     1 10.11.85.106         10.10.10.3    UP 00:34:32     D
```

show dmvpn

```
HUB1#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
==========================================================================

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     1 10.11.85.102         10.10.10.2    UP 00:34:25     D
     1 10.11.85.106         10.10.10.3    UP 00:34:32     D

HUB1# show interface Tunnel0
```

```
show interface Tunnel0
```

```
SPOKE1#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Description: DMVPN Fase3 - Spoke1
  Internet address is 10.10.10.2/27
  MTU 17912 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.11.85.102 (Ethernet0/0)
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with Ethernet0/0
          Set of tunnels with source Ethernet0/0, 1 member (includes iterators), on interface <OK>
  Tunnel protocol/transport multi-GRE/IP
    Key 0x14CC1, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1472 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPSec (profile "PROF-DMVPN-VTI")
  Last input 00:00:03, output 00:35:48, output hang never
```

```
show ip eigrp neighbors
```

```
HUB1# show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)
H   Address                 Interface              Hold Uptime   SRTT   RTO  Q  Seq
                                                   (sec)         (ms)       Cnt Num
1   10.10.10.2              Tu0                      11 00:35:19   20   1440  0  3
0   10.10.10.3              Tu0                      13 00:35:25   18   1440  0  4
HUB1#show ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.11.85.97 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
D        10.11.85.0/27 [90/26905600] via 10.10.10.2, 00:35:27, Tunnel0
D        10.11.85.32/27 [90/26905600] via 10.10.10.3, 00:35:34, Tunnel0
```

show ip route eigrp

```
HUB1#show ip route eigrp
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 10.11.85.97 to network 0.0.0.0

      10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
D        10.11.85.0/27 [90/26905600] via 10.10.10.2, 00:35:27, Tunnel0
D        10.11.85.32/27 [90/26905600] via 10.10.10.3, 00:35:34, Tunnel0
HUB1#
```

Comando trace, demostración nuevas entradas

```
PC5> trace 10.11.85.34
trace to 10.11.85.34, 8 hops max, press Ctrl+C to stop
 1   10.11.85.1   0.374 ms  0.171 ms  0.156 ms
 2   10.10.10.1   2.041 ms  0.406 ms  0.711 ms
 3   10.10.10.3   7.762 ms  1.805 ms  1.287 ms
 4   *10.11.85.34   3.101 ms (ICMP type:3, code:3, Destination port unreachable)

PC5>
```

Demostración del Shortcut

Paso	Acción

1	Antes: show ip route 10.11.85.0 en SPOKE1 → next-hop = HUB (10.10.10.1)
2	Después del tráfico sostenido: next-hop cambia a SPOKE1 (10.10.10.3)
3	show dmvpn en SPOKE2 mostrará una segunda entrada dinámica

```
SPOKE1#show ip nhrp
10.10.10.1/32 via 10.10.10.1
   Tunnel0 created 00:33:12, never expire
   Type: static, Flags: used
   NBMA address: 10.11.85.98
SPOKE1#show ip nhrp
10.10.10.1/32 via 10.10.10.1
   Tunnel0 created 00:33:34, never expire
   Type: static, Flags: used
   NBMA address: 10.11.85.98
10.10.10.3/32 via 10.10.10.3
   Tunnel0 created 00:00:10, expire 00:02:54
   Type: dynamic, Flags: used temporary
   NBMA address: 10.11.85.98
10.11.85.34/32
   Tunnel0 created 00:00:10, expire 00:02:54
   Type: incomplete, Flags: negative
   Cache hits: 2
```

```
Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

 # Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
 ----- --------------- --------------- ----- -------- -----
     3 10.11.85.98          10.10.10.1    UP 00:33:40     S
                            10.10.10.3    UP 00:01:17     D
     0 UNKNOWN             10.11.85.34  NHRP    never    IX
```